Part II: Eigenface Feature Detection

Background

Principle Component Analysis (PCA) detects the main patterns of can be used to detect facial features from a data set of random faces. Of the dataset, each face image was reshaped from a matrix into a long column vector, which were then placed into a large matrix A, in which each column represented features of one face.

The average of each column of A was then computed, giving the "mean" face.

$$A_{mean} = \frac{1}{N} \sum_{i=1}^{N} a_i$$

This was then subtracted from each image, producing a centred matrix (mean centring). Mean centring was important since PCA only investigates how each image differs from the average, while not necessarily considering the brightness or other common facial features.

$$A_{centered} = A - A_{mean}$$

For standard PCA the next step would involve finding the eigenvectors and eigenvalues of the data's covariance matrix, C:

$$C = \frac{1}{N} A A^T$$

Instead there is a simpler method to accomplish the needed output for the detector. Singular value decomposition (SVD), allows easy access to the eigenvectors necessary for feature detection, thus SVD was applied to the centred matrix A, given by:

$$A_{centered} = U \Sigma V^T$$

Each parts of the matrix of the SVD form represented different things:

- Columns of U represented the eigenfaces, which are main patterns of variations in the image set.
- The largest singular values correspond to the most important eigenfaces, capturing the strongest patterns PCA does not capture, such as: brightness, shadows and most importantly: facial structures, which is used for the detection of moustaches as required specifically for the task.

Implementation

After calculating mean face, centred matrix and doing the SVD, the eigenfaces are then extracted and presented in Figure one.

Figure 1: First 20 Eigenfaces visualised

Visual inspection tells us eigenface 13 is a good candidate to use for a moustache detector.

Taking eigenface 13 we calculate the orthogonal projection of each centered face onto the eigenface vector to get a “similarity/alignment” score which represents how similar the face is to the eigenface, by projecting it onto the subspace of the eigenface.

Now having the scores for all the faces by further visual inspection it is determined which faces have moustaches and which do not. Then by taking the averages of each set, the scores of faces with and without moustaches, a threshold can be calculated giving an appropriate division point between faces with and without moustaches such that a new face can be provided and sorted accordingly.

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Moustache detector Accuracy: 89.3%
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Figure 2: Accuracy percentage of moustache detector

Resulting in a fairly accurate moustache detector, figure two showing a 89.3% accuracy rate of detection.

Efficacy

Figure 3: 30 moustache detected faces

The system achieved an approximate accuracy of 89.3%, correctly classifying the majority of images with clear moustache features. In figure three, analysing the 30 random faces MATLAB used for moustache detection. The results appeared to be reasonable. With images that had a highly visible moustache, to the human eye, MATLAB seemed to have detected a moustache too. Indicating, as long as a moustache was visible, the detector would be able to recognise it too. However, the detector was not completely reliable, some images without moustaches were classified as having moustaches; likely due to strong shadows around the mouth or nose, indicating imperfection, suggesting detector was not detecting image patterns similar to moustache eigenfaces. These patterns included dark regions near the upper lip, shadows caused by lighting.

Figure 4: 13 altered faces tested with moustache detector

After applying the moustache detector on manipulated images, it was revealed that the detector performed significantly worse under extreme conditions (e.g. if the modifications were either too much or too little) and struggled to identify differences in moustache weight and types. This limitation is due to the detector relying on one eigenface to fully encode the moustache feature, along with the global nature of PCA as it cannot truly single out facial features. Figure four shows faces modified in numerous ways., notably faces with lighter-colored moustaches were not detected, this enforces the idea a single eigenface does not properly encode all types of moustache. Furthermore, subjects with thin or less prominent moustaches were incorrectly categorised, indicating that the eigenface may not be fully representative of all moustache variations. Conversely, these results might also reinforce the detector's robustness, as the manipulated images lacked the convincing quality required to accurately simulate the presence of moustaches.

In summary, the detector can be reliable provided that conditions are consistent between images and the feature being detected is also consistent from image to image. The detector is robust however prone to variations in camera angles, lighting, and facial hair characteristics, these all are liable to result in erroneous detections.

Conclusion

A technical report in PDF format of approximately **10 pages**. The report must contain an introduction,

separate sections for Parts I and II (details below), **conclusions**, and **references**.

Back to digital health now: the idea is that these -dimensional coordinate vectors could be used to train

a machine learning system to identify warning signs of disease. The "faces" are really going to be MRI

brain scans, remember, and the features will be whatever the SVD identifies as the most distinguishing

characteristics of human MRI brain scans. With a very large dataset of pre-labelled healthy and diseased

brain images, the system can try to learn which combinations of features are associated with greater

incidence of (say) neurodegenerative disease. New patients whose coordinate vectors have a lot in

common with these learned "biomarkers" could then benefit from much earlier intervention than would be

possible using only the diagnostic expertise of a human operator, with only a human lifetime's capacity for

learning.